



Student Name: _____

Teacher: _____

Sydney Technical High School

2023

YEAR 11
PRELIMINARY COURSE
EXAMINATION

Mathematics Extension 1

General Instructions

- Working Time – 1 hour and 30 minutes
- Write using black pen.
- Calculators approved by NESA may be used.
- A reference sheet is provided.
- For questions in Section II, show relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for careless work or illegible writing.
- **Begin each question on a new page.**

Total marks:
72

Section I — 6 marks

- Attempt Questions 1 – 6.
- Allow about 10 minutes for this part.

Section II — 66 marks

- Attempt Questions 7 – 12.
- Allow about 80 minutes for this section.

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SECTION I

6 marks

Attempt Questions 1-6.

Allow about 10 mins for this section.

Use the multiple-choice answer sheet for Questions 1-6.

1. At a family gathering, 6 adults and 3 children line up for a photo in 2 lines, children in the front and adults in the back. Which expression gives the number of possible arrangements?

- (A) $9!$
- (B) $3! \times 6!$
- (C) $3! \times 6! \times 2!$
- (D) $\frac{3! \times 6!}{2!}$

2. Given $f(x) = \sqrt{3 - \sqrt{x}}$, what is the range for $y = f^{-1}(x)$?

- (A) $(0,9]$
- (B) $(0,\sqrt{3})$
- (C) $[0,9]$
- (D) $[0,\sqrt{3}]$

3. How many terms will the expansion $(a + b)^{51} + (a - b)^{51}$ have?

- (A) 102
- (B) 25
- (C) 26
- (D) 0

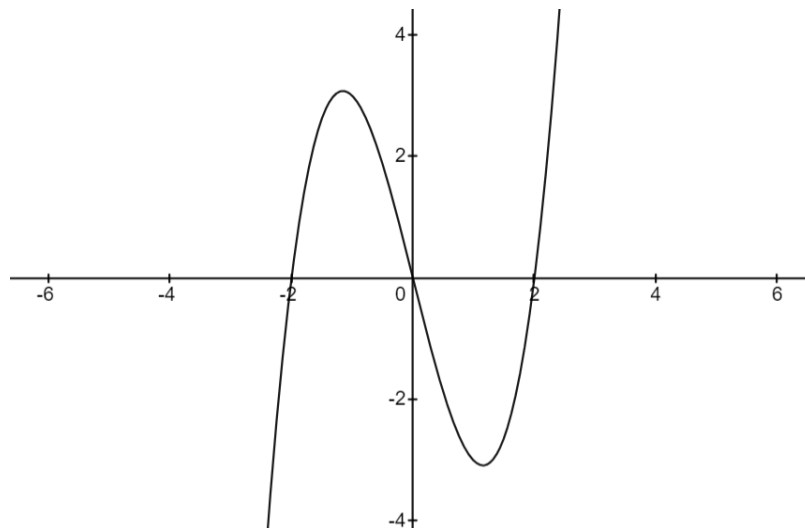
4. In a school with 1200 students, which of the following statements must be true?

- (A) At least 4 students will have the same birthday.
- (B) Exactly 4 students will have the same birthday.
- (C) There will be 4 birthdays every day.
- (D) There are at least 4 birthdays on each day of the year.

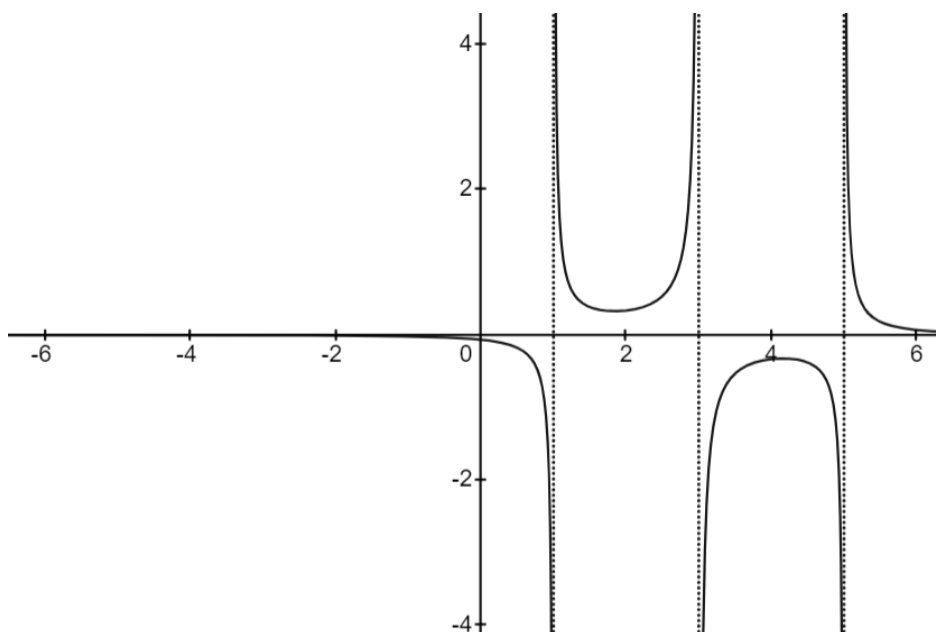
5. A curve is defined parametrically by $x = 4t^2$ and $y = 8t$. The curve will be a:

- (A) One-to-one function
- (B) Many-to-one function
- (C) One-to-many relation
- (D) Many-to-many relation

6. The graph of $y = f(x)$ is shown below.



Which transformation would result in the following function:



- (A) $y = \frac{1}{f(x)+3}$
- (B) $y = \frac{1}{f(x)-3}$
- (C) $y = \frac{1}{f(x-3)}$
- (D) $y = \frac{1}{f(x+3)}$

End of Section I

SECTION II

66 marks

Attempt Questions 7 – 12.

Allow about 80 minutes for this section.

Start your answer for each question on a new page in the answer booklet.

In Questions 7 – 12, your responses should include all relevant mathematical reasoning and/or calculations.

Question 7 (11 marks) Start a NEW page.	Marks
a) Solve the inequality $x(2 - x)(x + 1) \geq 0$	2
b) The polynomial $P(x) = x^3 - 6x^2 - 2x + c$ has a factor of $x - 2$. What is the value of c ?	2
c) Find the constant term in the expansion $\left(2x + \frac{1}{x}\right)^8$.	2
d) How many distinct arrangements of the letters of the word ALGEBRA are possible in a straight line if the A's are together?	2
e) Find the general solution to $2 \cos x = \sqrt{3}$. Leave your answer in terms of π .	3

End of Question 7

Question 8 (11 marks) Start a NEW page.	Marks
Consider the function $f(x) = \frac{2}{x+1}$ and its inverse function $f^{-1}(x)$.	
a) State the domain and range of $y = f(x)$	2
b) Evaluate $f^{-1}(4)$.	2
c) Find the equation of $f^{-1}(x)$.	2
d) For what values does $f(x) = f^{-1}(x)$.	2
e) Sketch $y = f^{-1}(x)$.	3

End of Question 8

Question 9 (11 marks) Start a NEW page.**Marks**

- a) Let α, β and γ be the roots of the equation $x^3 - 3x^2 - 6x - 1 = 0$.
- i. Find $2\alpha + 2\beta + 2\gamma$. **1**
- ii. Find $\alpha^2 + \beta^2 + \gamma^2$ **2**
- b) Given $A(x) = (x^2 - 1)Q(x) + (3x + 1)$, what is the remainder when $A(x)$ is divided by $x + 1$? **2**
- c) Solve the inequality $\frac{3x}{2x-1} \geq 1$ **3**
- d) The coefficients of x^2 and x^{-1} in the expansion of $\left(x - \frac{2a}{x^2}\right)^5$ are equal. **3**
- If a is non-zero find the value of a .

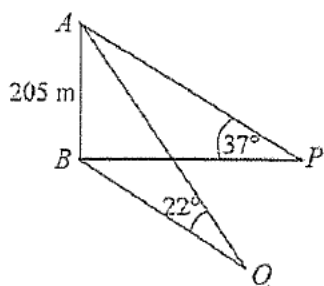
End of Question 9**Question 10** (11 marks) Start a NEW page.**Marks**

- a) By considering $(1 + x)^{15}$ show that:

2

$$\binom{15}{4} = \binom{8}{0}\binom{7}{4} + \binom{8}{1}\binom{7}{3} + \binom{8}{2}\binom{7}{2} + \binom{8}{3}\binom{7}{1} + \binom{8}{4}\binom{7}{0}$$

- b) A is 205 metres above the horizontal plane BPQ . AB is vertical. The angle of elevation of A from P is 37° and the angle of elevation of A from Q is 22° . P is due East of B and Q is south 47° east from B .



NOT TO SCALE

- i. State $\angle PBQ$ **1**
- ii. Calculate the distance from P to Q , to the nearest metre. **3**

- c) The polynomial $P(x) = 2x^3 - 12x^2 + k$ has a double root. **3**
 Given k is a positive integer, find its value.
- d) Show that $\frac{(x^2-4)!}{(x-2)(x^2-5)!} = x + 2$ **2**

End of Question 10

Question 11 (11 marks) Start a NEW page. **Marks**

- a) Consider $P(x) = x^3 - 5x^2 - 2x + 24$.
- i. Justify why $x - 4$ is a factor of $P(x)$. **1**
 - ii. Fully factor $P(x)$. **2**
 - iii. Hence sketch $y = P(x)$ **2**
 - iv. Hence sketch $y = \sqrt{P(x)}$ **2**
- b) From a group of 6 boys and 6 girls, 8 are chosen at random to form a group.
- i. How many different groups of 8 people can be formed? **1**
 - ii. How many of these groups will consist of 4 boys and 4 girls? **1**
 - iii. 4 boys and 4 girls are chosen and placed around a circle. **2**
 What is the probability that the circle has the boys and girls alternating?

End of Question 11

Question 12 (11 marks) Start a NEW page.

Marks

a) Consider $f(x) = 5 - |x + 2|$

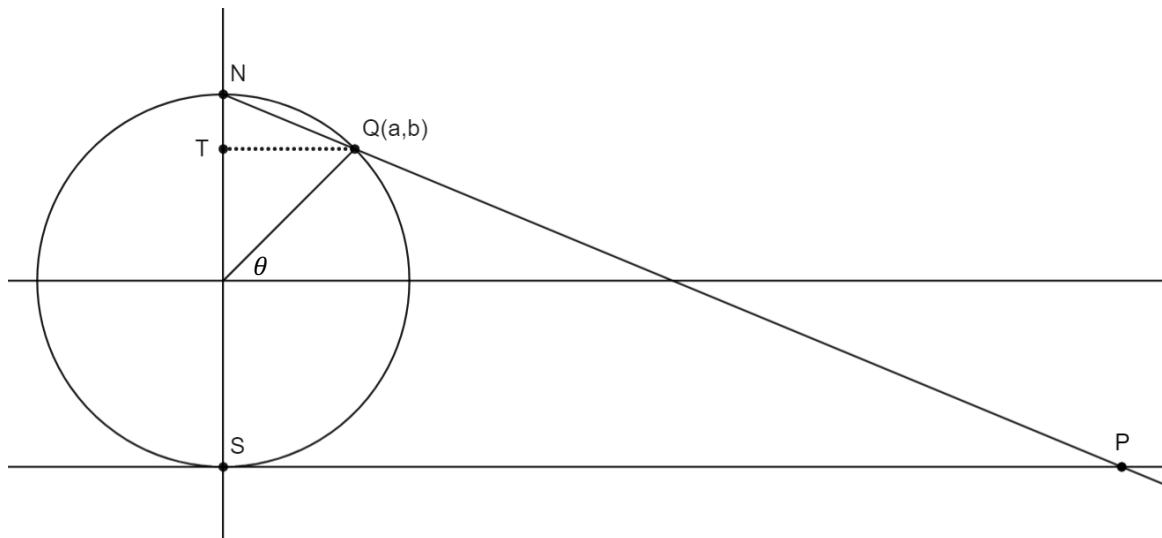
i. Graph $y = f(x)$

2

ii. Hence solve $|x + 2| + |x - 3| = 5$

1

b) In the diagram below, $Q(a, b)$ is a point on the unit circle $x^2 + y^2 = 1$ at an angle θ from the positive x - axis, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. The line through $N(0,1)$ and Q intersects the line $y = -1$ at P . The points $T(0, b)$ and $S(0, -1)$ are on the y - axis.



i. Using the fact that $\triangle TQN$ and $\triangle SPN$ are similar, show that

2

$$SP = \frac{2 \cos \theta}{1 - \sin \theta}$$

ii. Show that $\frac{\cos \theta}{1 - \sin \theta} = \sec \theta + \tan \theta$

2

iii. Given $SQ \perp PN$, explain why $\angle SNP = \frac{\theta}{2} + \frac{\pi}{4}$

2

iv. Hence, show that $\tan\left(\frac{\theta}{2} + \frac{\pi}{4}\right) = \sec \theta + \tan \theta$

2

End of Exam

Section 1 - MC

1. B

2. C

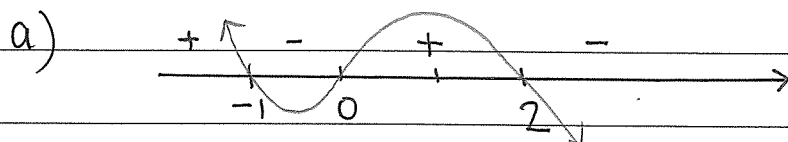
3. C

4. A

5. C

6. C

Question 7



$$x \leq -1 \cup 0 \leq x \leq 2$$

b) $P(2) = 0$

$$0 = 2^3 - 6 \times 2^2 - 2 \times 2 + c$$

$$= -20 + c$$

$$c = 20$$

c) $T_{k+1} = {}^8C_k (2x)^{8-k} (x^{-1})^k$

$$\therefore k = 4$$

$$T_5 = {}^8C_4 (2)^4$$

$$= 1120$$

d) AA L G E B R $6! = 720$

e) $2 \cos x = \sqrt{3}$

$$\cos x = \frac{\sqrt{3}}{2}$$

$$x = \frac{\pi}{6} \pm 2\pi n, n \in \mathbb{Z} \quad \text{or} \quad \frac{11\pi}{6} \pm 2\pi n, n \in \mathbb{Z}$$

Question 8

a) $D: x \in \mathbb{R}, x \neq -1$

$R: y \in \mathbb{R}, y \neq 0$

b) Solve:

$$4 = \frac{2}{x+1}$$

$$4x+4=2$$

$$4x = \frac{2}{4}$$

$$x = -\frac{1}{2}$$

$$\therefore f^{-1}(4) = -\frac{1}{2}$$

c) $x = \frac{2}{y+1}$

$$(y+1)x = 2$$

$$y+1 = \frac{2}{x}$$

$$y = \frac{2}{x} - 1$$

d) $f(x) = f^{-1}(x)$ at the $y=x$ line.

$$\therefore x = \frac{2}{x+1}$$

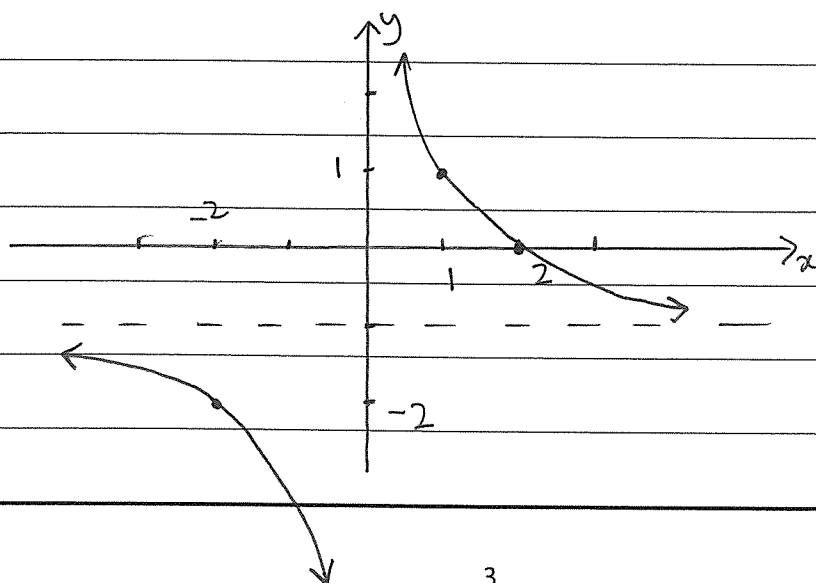
$$x(x+1) = 2$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2, 1$$

e)



Question 9

a) i. $2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma)$

$$= 2 \times 3$$

$$= 6$$

ii. $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$= (3)^2 - 2(-6)$$

$$= 21$$

b) $R(x) = 3x + 1$

$$R(-1) = -2$$

c) $\frac{3x}{2x-1} \geq 1 \quad x \neq \frac{1}{2}$

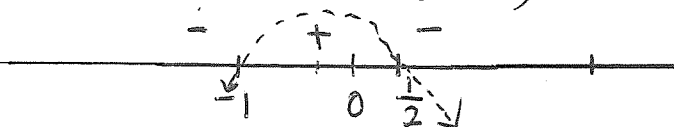
$$3x(2x-1) \geq (2x-1)^2$$

$$0 \geq (2x-1)^2 - 3x(2x-1)$$

$$0 \geq (2x-1)(2x-1-3x)$$

$$0 \geq (2x-1)(-x-1)$$

$$0 \geq -(2x-1)(x+1)$$



$$\therefore x \leq -1 \cup x > \frac{1}{2}$$

d) $T_{k+1} = {}^5C_k (x)^{5-k} \left(-\frac{2a}{x^2}\right)^k$

$$= {}^5C_k x^{5-k} (-2ax^{-2})^k$$

For x^2 , $k=1$ and x^{-1} , $k=2$

$$\therefore {}^5C_1 (-2a)^1 = {}^5C_2 (-2a)^2$$

$$-10a = 40a^2$$

$$0 = 40a^2 + 10a, \quad a \neq 0$$

$$\therefore a = -\frac{1}{4}$$

Question 10

a) $(1+x)^{15} = (1+x)^8(1+x)^7$

Consider x^4 terms.

For $(1+x)^{15}$, ${}^{15}C_4 x^4$

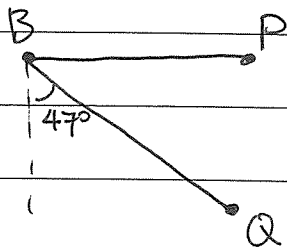
For $(1+x)^8(1+x)^7$

$${}^8C_0 x^7 {}^7C_4 x^4 + {}^8C_1 x^7 {}^7C_3 x^3 + {}^8C_2 x^2 {}^7C_2 x^2 + {}^8C_3 x^3 {}^7C_1 x + {}^8C_4 x^4 {}^7C_0$$

Equating coefficients:

$${}^{15}C_4 = {}^8C_0 {}^7C_4 + {}^8C_1 {}^7C_3 + {}^8C_2 {}^7C_2 + {}^8C_3 {}^7C_1 + {}^8C_4 {}^7C_0$$

b) i.



$$\angle PBQ = 43^\circ$$

ii. $\frac{BQ}{205} = \tan 68^\circ$

$$BQ = 205 \tan 68^\circ$$

$$\frac{BP}{205} = \tan 53^\circ$$

$$BP = 205 \tan 53^\circ$$

$$\begin{aligned} PQ^2 &= (205 \tan 68^\circ)^2 + (205 \tan 53^\circ)^2 - 2(205^2) \tan 68^\circ \tan 53^\circ \cos 43^\circ \\ &= 129.553.2221 \dots \end{aligned}$$

$$PQ \doteq 360 \text{ m (nearest m)}$$

c) $P'(x) = 6x^2 - 24x$

$$P'(x) = 0 \text{ for double root}$$

$$\text{i.e. } 6x^2 - 24x = 0$$

$$6x(x-4) = 0 \quad \text{since } k \neq 0, \text{ double root can't be } 0$$

$\therefore$ Double root is $x=4$

$$P(4)=0$$

$$0 = 2 \times 4^3 - 12 \times 4^2 + k$$

$$0 = -64 + k$$

$$k = 64$$

$$\begin{aligned} \text{d) } \frac{(x^2-4)!}{(x-2)(x^2-5)!} &= \frac{(x^2-5)!(x^2-4)}{(x-2)(x^2-5)!} \\ &= \frac{(x-2)(x+2)}{(x-2)} \\ &= x+2 \end{aligned}$$

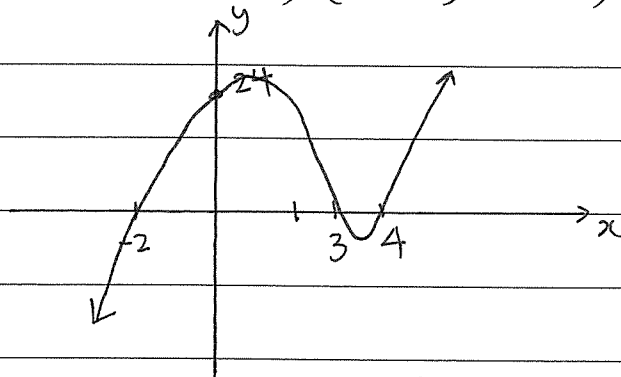
Question 11

$$\begin{aligned} \text{a) i. } P(4) &= 4^3 - 5 \times 4^2 - 2 \times 4 + 24 \\ &= 0 \end{aligned}$$

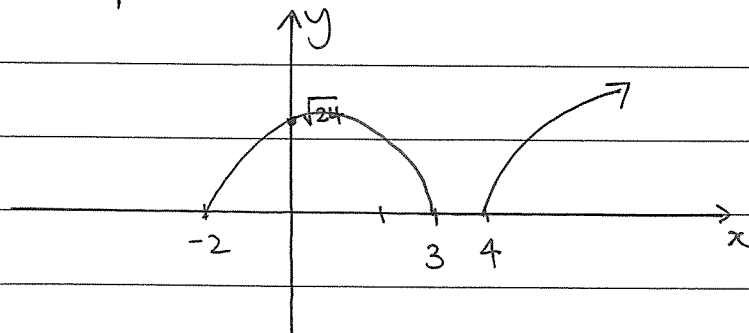
Since $P(4)=0$, $x-4$ is a factor of $P(x)$

$$\begin{aligned} \text{ii. } P(x) &= (x-4)(x^2-x-6) \text{ By inspection} \\ &= (x-4)(x-3)(x+2) \end{aligned}$$

iii.



iv.



b) i. ${}^{12}C_8 = 495$

ii. ${}^6C_4 \times {}^6C_4 = 225$

iii. No restrictions: $(8-1)!$

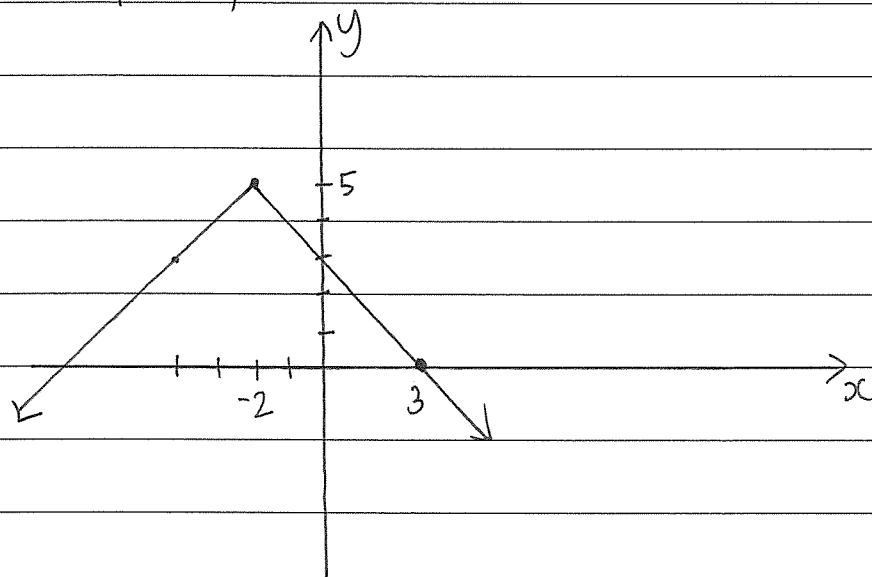
Alternating: $(4-1)! \times 4!$

$$P(\text{alternating}) = \frac{3! \times 4!}{7!}$$

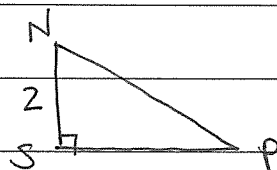
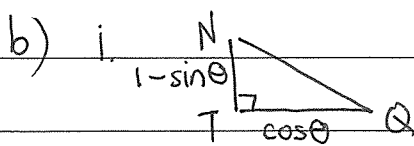
$$= \frac{1}{35}$$

Question 12

a) i. $y = 5 - |x+2|$



ii. $-2 \leq x \leq 3$

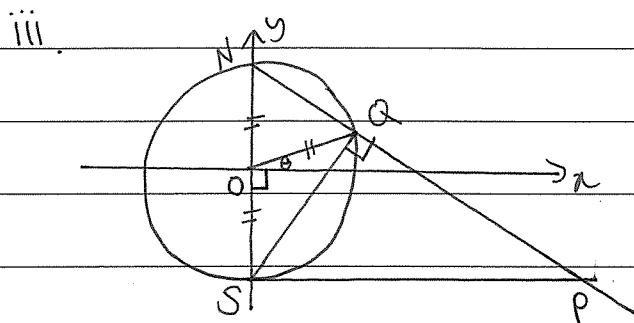


$$\frac{1 - \sin \theta}{\cos \theta} = \frac{2}{p}$$

$$p(1 - \sin \theta) = 2 \cos \theta$$

$$p = \frac{2 \cos \theta}{1 - \sin \theta}$$

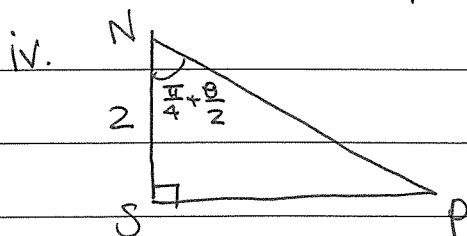
$$\begin{aligned}
 \text{ii. } \frac{\cos \theta}{1 - \sin \theta} &= \frac{\cos \theta}{1 - \sin \theta} \times \frac{1 + \sin \theta}{1 + \sin \theta} \\
 &= \frac{\cos \theta (1 + \sin \theta)}{1 - \sin^2 \theta} \\
 &= \frac{\cos \theta (1 + \sin \theta)}{\cos^2 \theta} \\
 &= \frac{1 + \sin \theta}{\cos \theta} \\
 &= \sec \theta + \tan \theta
 \end{aligned}$$



$$\begin{aligned}
 \angle NOQ &= \pi - \left(\frac{\pi}{2} + \theta \right) \quad (\text{angle sum of straight line}) \\
 &= \frac{\pi}{2} - \theta
 \end{aligned}$$

$$\begin{aligned}
 \angle SNP &= \angle OQN \quad (\text{equal angles opposite equal sides}) \\
 \frac{\pi}{2} - \theta + 2\angle SNP &= \pi \quad (\text{angle sum of } \triangle ONQ)
 \end{aligned}$$

$$\begin{aligned}
 2\angle SNP &= \frac{\pi}{2} + \theta \\
 \angle SNP &= \frac{\pi}{4} + \frac{\theta}{2}
 \end{aligned}$$



$$\begin{aligned}
 \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) &= \frac{SP}{2} \\
 &= \frac{\frac{2\cos \theta}{1 - \sin \theta}}{2} \quad (\text{from part i}) \\
 &= \frac{\cos \theta}{1 - \sin \theta} \\
 &= \sec \theta + \tan \theta \quad (\text{from part ii})
 \end{aligned}$$